Usability Testing Tool

YouTube Link: https://youtu.be/StLD39ZZgck

Overview

The Usability Testing Tool was developed using Streamlit to conduct structured usability evaluations. The system guides participants through a series of pages where they provide consent, demographic information, complete tasks, and give feedback about their experience. All responses are stored in CSV files and summarized in a report page using tables and charts generated with Altair and analyzed with Pandas.

Feature	Description	Data Stored
Introductory Page	Explains the purpose of the study and instructions	No
Consent Form	Participants must agree before continuing	No
Demographics	Collects participant background information	responses.csv
Tasks Page	Participants perform usability tasks with a timer	task_data.csv
Exit Questionnaire	Participants rate their experience and provide feedback	exit_questionnaire.csv
Report Page	Displays collected results using tables and charts	Reads all CSV files

Workflow:

The application follows a linear workflow to guide participants through the usability study. Users cannot skip steps, ensuring that all required information is collected before moving forward.

The process begins on the **Introductory Page**, where participants are provided with information about the study and an overview of the application structure. From this page, users can only proceed to the next step, creating a controlled flow similar to a form.

After continuing, participants reach the **Consent Form Page**, where they must read and agree to the terms of participation. If the consent checkbox is not selected, the system displays a warning message and prevents the user from advancing.

Next, participants complete the **Demographics Page**, where they provide information such as age range, gender, education level, and technology experience. These responses are saved in a CSV file so they can later be analyzed and visualized in the report.

Participants then proceed to the **Tasks Page**, where they select one of several usability tasks. A timer can only start once a task is selected. When the task begins, the timer records the completion time. After finishing, users choose either **"**Finish**"** or **"**Could Not Finish," both of which stop the timer. Participants may also enter notes describing their experience or any issues encountered. Task outcomes, completion times, and comments are stored in a CSV file.

The next step is the **Exit Questionnaire**, where users rate their overall experience and the ease of completing tasks. They can also provide written feedback about what they liked and what could be improved.

Finally, the **Report Page** summarizes all collected data. The information is displayed using tables and charts that highlight key usability metrics such as task success rate, average completion time, and user satisfaction ratings. This summary allows developers to quickly identify usability strengths and areas that may require improvement.